

Ice-column damage evolution from van der Veen crevasse mechanics

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Abstract

Surface and basal crevasses determine whether an ice column retains a connected load-bearing ligament. The van der Veen models predict whether an existing crevasse propagates or arrests, but they do not specify a propagation rate. This distinction prevents their direct use as a prognostic damage equation. Here, the fractured thickness fraction is written as $D = D_s + D_b$, and the van der Veen stress-intensity factors are retained as the fracture calculation. A single, explicit kinetic assumption then converts fracture excess into crack-tip speed. The resulting law predicts the same arrest depths as the original criterion and declares full-thickness failure when $D = 1$. The crack-speed scale is new physics, not a result of the static criterion, and requires calibration against time-resolved crevasse growth.

1 Mechanical question

Crevasse-depth calculations provide a geometric measure of ice damage, but a static depth is not an evolution law. In particular, the linear elastic fracture mechanics (LEFM) models of [van der Veen \(1998b\)](#) and [van der Veen \(1998a\)](#) calculate the mode-I stress intensity factor at an assumed crack tip. They predict propagation when this factor exceeds the fracture toughness of ice. They do not calculate DD/Dt .

This note makes the smallest additional assumption needed to obtain that rate. The calculation is separated into two parts. First, van der Veen mechanics determines whether each crack tip is above or below its propagation threshold. Second, an explicitly assumed kinetic function maps the threshold excess to a crack-tip speed. A reduced zero-stress version is also given because it leads to an algebraic right-hand side suitable for a depth-integrated ice model. The reduced version is a separate approximation and is not the van der Veen LEFM calculation.

2 Geometry, boundary conditions, and assumptions

Consider an ice column of thickness H measured in meters. A surface crevasse penetrates downward a distance d_s , and a basal crevasse penetrates upward a distance d_b . The state variables are

$$D_s = \frac{d_s}{H}, \quad D_b = \frac{d_b}{H}, \quad D = D_s + D_b. \quad (1)$$

The model applies while $0 \leq D < 1$. Full-thickness connection occurs at the first-hitting time

$$t_f = \min\{t : D(t) = 1\}. \quad (2)$$

At t_f , the column is declared fractured through its thickness; the damage equation is not continued beyond this event.

Table 1 summarizes the notation used below.

The calculation makes five assumptions.

Table 1: Notation. Subscripts s, b, i, and m denote surface, basal, ice, and surface meltwater quantities, respectively.

Symbol	Meaning and SI units	Symbol	Meaning and SI units
H	Ice thickness (m)	d_s, d_b	Surface and basal penetration depths (m)
D_s, D_b	Surface and basal fractured fractions (dimensionless)	D	Total fractured fraction, $D_s + D_b$ (dimensionless)
$\mathbf{u}$	Horizontal ice velocity (m s^{-1})	$\mathcal{D}_t$	Material derivative (s^{-1} when applied to D)
R_n	Resistive stress normal to a crevasse (Pa)	τ_{nn}, τ_{tt}	Deviatoric stresses normal and parallel to a crevasse (Pa)
ρ_i	Ice density (kg m^{-3})	ρ_m, ρ_b	Surface and basal water densities (kg m^{-3})
g	Gravitational acceleration (m s^{-2})	a_s, a_b	LEFM surface and basal crack lengths (m)
h_s	Surface water-column height (m)	δ_s	Dry length above water in a surface crack (m)
p_b^0	Basal water pressure at the crack mouth (Pa)	K_I^s, K_I^b	Mode-I stress intensity factors ($\text{Pa m}^{1/2}$)
K_{Ic}	Ice fracture toughness ($\text{Pa m}^{1/2}$)	F, G	Finite-slab weight functions (dimensionless)
λ, γ	Normalized crack length and integration coordinate (dimensionless)	v_s, v_b	Characteristic crack speeds (m s^{-1})
$\mathcal{F}_j$	Assumed kinetic function (dimensionless)	Σ	Resistive stress normalized by ice overburden (dimensionless)
W_s	Surface water-column fraction (dimensionless)	r_s, r_b	Water-to-ice density ratios (dimensionless)
H_{ab}	Thickness above basal water-pressure support (m)	A_b	H_{ab}/H (dimensionless)
Ψ_s, Ψ_b	Reduced normalized opening stresses (dimensionless)	t_f	First time at which $D = 1$ (s)

1. Surface and basal crevasses are pre-existing mode-I cracks. The model predicts propagation, not nucleation, and does not include healing.
2. The tensile resistive stress R_n , measured in pascals, is uniform with depth. Ice overburden and water pressure are hydrostatic.
3. Surface water-column height and basal water pressure are prescribed by observations or a separate hydrology model.
4. Surface and basal cracks do not mechanically interact before $D = 1$. This assumption isolates their separate contributions but becomes least accurate when the remaining ligament is short.
5. The dimensionless fractions are advected with the ice. Crack-tip speed is measured relative to affine vertical deformation of the column.

With horizontal ice velocity $\mathbf{u}$, the material derivative is

$$\mathcal{D}_t(\cdot) = \frac{\partial(\cdot)}{\partial t} + \mathbf{u} \cdot \nabla(\cdot). \quad (3)$$

The resistive stress should be calculated normal to the crevasse. In a three-dimensional shallow model this gives $R_n = 2\tau_{nn} + \tau_{tt}$, where t is parallel to the crevasse; using τ_{nn} alone is not mechanically equivalent (Reynolds et al., 2025).

3 van der Veen fracture calculation

The van der Veen criterion is

$$K_I^j > K_{Ic}, \quad j \in \{s, b\}, \quad (4)$$

where K_I^j and K_{Ic} have units of $\text{Pa m}^{1/2}$. A crack propagates while equation (4) holds and arrests at the first stable depth where $K_I^j = K_{Ic}$.

For a surface crack, let $a_s = HD_s$, let y increase downward from the ice surface, and let δ_s be the dry crack length above the water surface. The water-column height is $h_s = \langle a_s - \delta_s \rangle_+$. The surface stress intensity factor is

$$\begin{aligned} K_I^s = & F(\lambda_s) R_n \sqrt{\pi a_s} - \frac{2\rho_i g}{\sqrt{\pi a_s}} \int_0^{a_s} y G\left(\lambda_s, \frac{y}{a_s}\right) dy \\ & + \frac{2\rho_m g}{\sqrt{\pi a_s}} \int_0^{a_s} \langle y - \delta_s \rangle_+ G\left(\lambda_s, \frac{y}{a_s}\right) dy, \quad \lambda_s = \frac{a_s}{H}. \end{aligned} \quad (5)$$

The three terms are opening by resistive stress, closure by ice overburden, and opening by meltwater pressure.

For a basal crack, let $a_b = HD_b$, let y increase upward from the bed, let p_b^0 be water pressure at the crack mouth, and let ρ_b be the density of water in the crack. The basal stress intensity factor is

$$K_I^b = \frac{2}{\sqrt{\pi a_b}} \int_0^{a_b} \left\{ R_n - \rho_i g(H - y) + \langle p_b^0 - \rho_b g y \rangle_+ \right\} G\left(\lambda_b, \frac{y}{a_b}\right) dy, \quad \lambda_b = \frac{a_b}{H}. \quad (6)$$

Here ice overburden closes the crack, whereas basal water pressure and tensile resistive stress open it. Appendix A gives the finite-slab weight functions F and G .

LEFM requires a finite initial flaw because K_I^j is not defined at zero crack length. The initial condition must therefore specify $D_j(0) = D_{j0} > 0$. In addition, the finite-slab weight functions

assume a rectangular body with a single edge crack. Grounded basal boundary conditions can change the calculated stress intensity factor, particularly for deep cracks (Jiménez and Duddu, 2018).

4 Damage evolution

The static criterion fixes the sign of the crack speed but not its magnitude. Write the unknown dimensionless kinetic relation as

$$\mathcal{F}_j(q) = 0 \quad \text{for } q \leq 1, \quad \mathcal{F}_j(q) > 0 \quad \text{for } q > 1. \quad (7)$$

If v_j is a characteristic crack speed in m s^{-1} , a local rate law consistent with the criterion is

$$\mathcal{D}_t D_s = \frac{v_s}{H} \mathcal{F}_s \left(\frac{K_I^s}{K_{Ic}} \right), \quad (8)$$

$$\mathcal{D}_t D_b = \frac{v_b}{H} \mathcal{F}_b \left(\frac{K_I^b}{K_{Ic}} \right). \quad (9)$$

These equations have units of inverse time. They also retain the original arrest condition because both rates vanish when $K_I^j \leq K_{Ic}$.

A minimal closure is a linear positive-part relation,

$$\mathcal{F}_j(q) = \langle q - 1 \rangle_+. \quad (10)$$

This relation is an assumption. It is not derived by van der Veen, and the calculation below should not be interpreted as a measured dynamic fracture law. With equation (10), the requested right-hand side is

$$\mathcal{D}_t D = \frac{v_s}{H} \left\langle \frac{K_I^s(D_s)}{K_{Ic}} - 1 \right\rangle_+ + \frac{v_b}{H} \left\langle \frac{K_I^b(D_b)}{K_{Ic}} - 1 \right\rangle_+, \quad D < 1. \quad (11)$$

Equation (11), together with the two component equations, is the prognostic van der Veen damage model. The calculation retains the original propagation threshold; the new parameters v_s and v_b set only the transient rate.

5 Reduced algebraic form

Some depth-integrated models do not evaluate stress-intensity factors. A separate zero-stress approximation places each crack tip where its net opening stress vanishes (Nye, 1955; Nick et al., 2010). Define

$$\Sigma = \frac{R_n}{\rho_i g H}, \quad W_s = \frac{h_s}{H}, \quad r_s = \frac{\rho_m}{\rho_i}, \quad r_b = \frac{\rho_b}{\rho_i}, \quad (12)$$

and

$$A_b = \frac{H_{ab}}{H}, \quad H_{ab} = H - \frac{p_b^0}{\rho_i g}. \quad (13)$$

Here Σ is tensile stress normalized by ice overburden, W_s is fractional surface-water depth, and A_b measures the ice thickness not supported by basal water pressure. Assuming water reaches the basal crack tip, the normalized opening stresses are

$$\Psi_s = \Sigma + r_s W_s - D_s, \quad (14)$$

$$\Psi_b = \Sigma - A_b - (r_b - 1) D_b. \quad (15)$$

Positive Ψ_j opens the corresponding crack. Applying the same minimal kinetic assumption gives

$$\boxed{\mathcal{D}_t D = \frac{v_s}{H} \langle \Sigma + r_s W_s - D_s \rangle_+ + \frac{v_b}{H} \langle \Sigma - A_b - (r_b - 1) D_b \rangle_+, \quad D < 1.} \quad (16)$$

The two component equations are obtained by retaining the corresponding term on the right-hand side.

The static limits provide a direct check. A dry surface crack arrests at $D_s^* = \Sigma$. A water-filled basal crack arrests at

$$D_b^* = \frac{1}{r_b - 1} \langle \Sigma - A_b \rangle_+, \quad (17)$$

provided $\rho_b > \rho_i$. These are the standard zero-stress crevasse depths. In contrast, a fully water-filled surface crack has $W_s = D_s$, and therefore

$$\Psi_s = \Sigma + (r_s - 1) D_s. \quad (18)$$

Freshwater is denser than ice, so $r_s > 1$. Once this crack propagates, equation (18) does not admit a positive arrest depth; the reduced model predicts full-thickness hydrofracture if the water column remains supplied.

6 Predictions and limitations

No observations are analyzed in this note. The model nevertheless makes three testable predictions.

First, measured arrest depths should satisfy $K_I^j = K_{Ic}$ when the stress, thickness, and water-pressure boundary conditions are inserted into equations (5) and (6). The reduced model instead predicts the algebraic depths above. Radar measurements of basal crevasse height and repeated surface topography could distinguish these calculations where the required boundary conditions are known.

Second, if v_j is constant, the assumed linear kinetics predicts

$$\frac{H \mathcal{D}_t D_j}{\langle K_I^j / K_{Ic} - 1 \rangle_+} = v_j \quad (19)$$

during propagation. Thus repeated measurements of crevasse depth should collapse to a constant speed for a given crack family. A systematic dependence of the left-hand side on fracture excess would reject the linear closure while leaving the van der Veen threshold calculation intact.

Third, for $0 < \Sigma < 1$, the reduced model predicts that a dry surface crack arrests at $D_s = \Sigma$, whereas a fully water-filled surface crack propagates through the thickness under the same tensile stress. This prediction is conditional on continued water supply. A hydrology model is required to determine whether that boundary condition persists.

The most important uncertainty is the kinetic function $\mathcal{F}_j$. Crack interaction, firm structure, spatially varying fracture toughness, and geometry-dependent weight functions are also omitted. These omissions affect the transient speed and may shift the arrest depth. They should be added only when observations require them.

7 Conclusion

This note rewrites the van der Veen propagation criterion as an evolution law for the fractured thickness fraction. The central result is equation (11): surface and basal damage grow only when their

stress-intensity factors exceed ice fracture toughness, and their rates add until $D = 1$. The threshold and arrest calculation come from van der Veen mechanics. The linear relation between fracture excess and crack speed is the sole new rate closure. Time-resolved measurements of crevasse growth could constrain that relation or replace it without changing the underlying fracture calculation.

A Finite-slab weight functions

The van der Veen finite-slab functions, as reproduced by Jiménez and Duddu (2018), are

$$F(\lambda) = 1.12 - 0.23\lambda + 10.55\lambda^2 - 21.72\lambda^3 + 30.39\lambda^4, \quad (20)$$

and

$$G(\lambda, \gamma) = \frac{3.52(1 - \gamma)}{(1 - \lambda)^{3/2}} - \frac{4.35 - 5.28\gamma}{(1 - \lambda)^{1/2}} + \left[\frac{1.30 - 0.30\gamma^{3/2}}{(1 - \gamma^2)^{1/2}} + 0.83 - 1.76\gamma \right] [1 - (1 - \gamma)\lambda], \quad (21)$$

where $\lambda = a/H$ and $\gamma = y/a$. The crack-tip singularity is integrable. Numerical quadrature should stop short of $\lambda = 1$, where the ice column has already met the full-thickness failure condition.

References

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